Yukin Pan

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Education

The Ohio State University (08/2017 - 05/2019)

- M.S. in Electrical and Computer Engineering.
- Coursework in Computer Vision, Machine Learning and Image Processing.

Wuhan University (09/2013 - 06/2017)

- B.S. in Automation, GPA 85/100 [Top 15%]
- AAA Graduate Student

Scholarships and Awards

- 2018: AI Star Intern in the JD AI Research [10/50]
- 2018: Ranked 2nd in the CVPR 2018 Workshop, LIP Contest
- 2017: AAA Graduate Student [10/450]
- 2016: Interdisciplinary Contest in Modeling (Honorable Mention)
- 2015: WHU Outstanding Student Leader [2/60]
- 2014: Avery Dennison Foundation Spirit of Invention Scholarship [10/150]

Professional Experience

AI Engineer, HomeTeam Live (03/2026 - present)

> AI-Powered Sports Video Broadcasting:

- Building AI-driven video broadcasting for live sports.

CV Engineer, vivo AI Research (07/2019 - 06/2021)

> Face Clustering in Digital Album:

- **Facial Feature Extraction**: Built a CNN tailored to extracting human facial features.
- **Graph Clustering Algorithm (GCA)**: Designed a clustering algorithm grounded in graph structure and K-Nearest Neighbors (KNN) theory.
- **On-Device Implementation**: Paired GCA with cosine distance metrics for real-time facial clustering on smartphones.
- **Results**: Raised Accuracy/Recall from 0.92/0.79 without GCA to 0.99/0.94 with it.
- **Deployment and Scale**: Shipped the algorithmic module on over 1 million vivo smartphones.

> Real-Time Human Segmentation on Mobile Devices:

- **CNN Construction**: An Encoder-Decoder model built on UNet and Depthwise Separable Convolution, implemented in TensorFlow.
- **Data Augmentation**: Grew the PFCN dataset (18,698 images) and the vivo dataset (5,000 images) past 60,000 images via cropping, flipping, shifting, zooming and alpha blending.
- **Inter-Frame Consistency**: Introduced alpha blending during processing to keep masks stable frame to frame.
- **Computational Efficiency**: Distillation and pruning took inference from 25 fps to 75 fps on Qualcomm Snapdragon 855 platforms.

> Real-Time Frame Interpolation and Motion Deblur on Mobile Devices:

- **CNN Construction**: A two-stage model rooted in the Super SloMo architecture, covering optical flow prediction and frame restoration.
- **Computational Efficiency**: Cut model FLOPs substantially through distillation and pruning.
- **Unsupervised Learning Strategy**: Proposed an unsupervised scheme enabling training on Adobe240fps and UCF101 datasets.
- **Parallel Computing Pipeline Built**: Engineered a pipeline with OpenCL on GPU and VCap on NPU, taking throughput from 7 fps to 40 fps.

Research Experience

AI Researcher, KUANGJINGBOXUAN (07/2021 - 10/2024)

> Real-Time Video Quality Enhancement for Mobile Devices:

- **Proposed a CNN Based Model**: Combined an attention module, skip connections, residual modules and mask modules to restore video quality in real time.
- **Optimized Computational Efficiency**: From 330 GOPs to 12 GOPs by implementing **Space2Depth**, **Distillation** and **Pruning**. Achieved inference time reduction from 100+ ms to 15 ms on the Qualcomm Snapdragon 8475 platform.
- **Noise Data Collection**: Devised an approach to capture noise from real-world footage, improving training-set realism and model robustness.
- **Data Augmentation Techniques**: ISP simulation, brightness adjustment, motion simulation and random disturbance to widen the model's operating range.
- **Loss Function Design**: Formulated gradient loss, edge loss and mask loss to keep performance stable under difficult capture conditions.
- **Deployment**: Built a pipeline with pre-processing and post-processing on GPU and model inference on NPU. Integrated this algorithmic module into the XIAOMI 12 Pro smartphone.

> Real-World Blind Super-Resolution via Feature Matching:

- **Constructed a VQGAN Based Model**: Incorporated Residual Blocks, Swin Transformer Blocks, a Codebook and a Feature Matching Module to blindly enhance video resolution from 720p to 1440p.
- **HR Dataset Build**: Built and augmented a high-resolution (HR) dataset from DIV2K, Flickr2K, DIV8K and real-world datasets via texture filtering.
- **Data Enhancement**: Used pretrained models FeMaSR, Real-ESRGAN and BasicVSR++ to further enhance high-frequency textures via image filtering and patch filtering.
- **Loss Function Design**: Formulated L2 loss, perceptual loss, GAN loss and codebook loss for robust model performance in the real world.
- **Model Pruning**: Pruned the model from 1600G FLOPs to 320G FLOPs (input shape 512x512x3) with almost the same performance.

Research Intern, JD AI Research, *Advisor: Prof. Tao Mei* (05/2018 - 09/2018)

> CVPR 2018 Workshop, LIP Contest (Ranked 2nd)

- **Data Augmentation**: Extended the training set with image rotations and reversals, plus Mask-RCNN for multi-human object detection.
- **Model Fine-Tuning and Fusion**: Customized and fine-tuned JPP-Net, SS-NAN, DenseNet, RefineNet and UPerNet on the LIP training set. Employed fusion strategies and integrated key-point data for performance refinement.
- **Hard Class Mining and Performance Optimization**: Identified and targeted several hard-to-classify categories, elevating overall model performance to 87.42% pixel accuracy.

Academic Publication

- Philippe Zhang, et al. "*Patch Progression Masked Autoencoder with Fusion CNN Network for Classifying Evolution Between Two Pairs of 2D OCT Slices*", International Challenge on Device-Independent Diabetic Macular Edema Onset Prediction. Springer Nature Switzerland, 2024. 97-105.
- Philippe Zhang, et al. "*Automatic Prediction of Center-Involved Diabetic Macular Edema Using Ultra-Widefield Color Fundus Photography*", International Challenge on Device-Independent Diabetic Macular Edema Onset Prediction. Springer Nature Switzerland, 2024. 20-29.
- Yunhan Huang, Li Ding, Yun Feng, Jiangtian Pan. "*Epidemic Spreading in Random Walkers with Heterogeneous Interaction Radius*", JSTAT 2016(10), 103501.

Skills

Programming: Python, Java, MATLAB, C++, OpenCL, CUDA

Deep Learning Framework: MegEngine, PyTorch, TensorFlow, TFLite, ONNX

Computer Vision Tools: OpenCV, Numpy, BasicSR

Professional Fields: Low-Level Vision, Image/Video Quality Enhancement, Real-Time CNN for Mobile Device